

Modem factory test

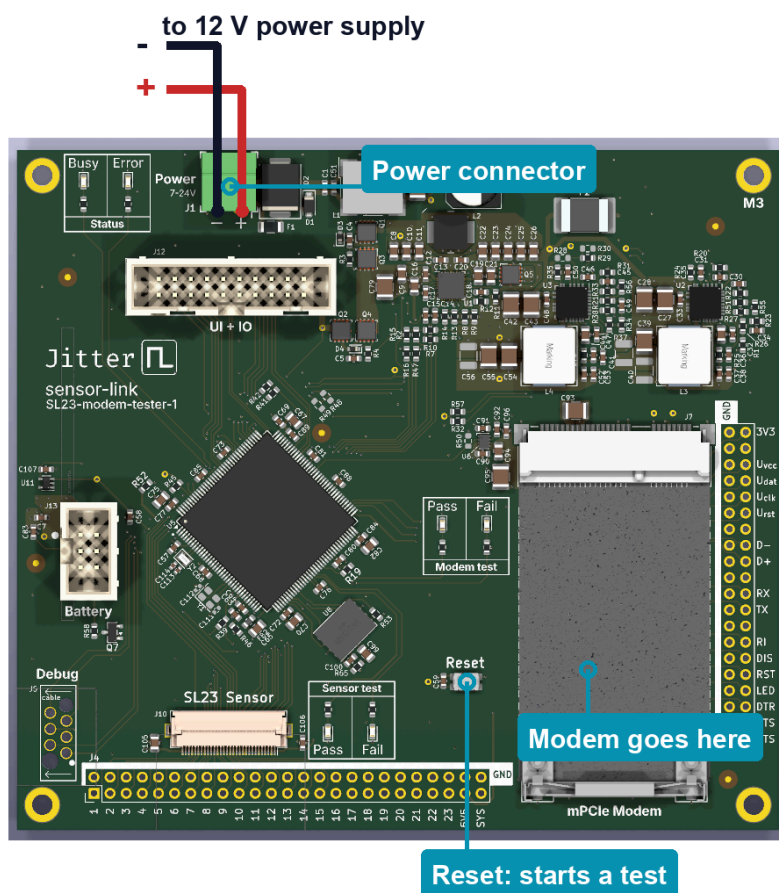
Work instruction · LTE module on its mini-PCle carrier · about 20 seconds per module

1. Equipment

Item	Setting
Power supply via Würth 691361300002 or 691368300002B	12.0 V, 1 A or more
SWD programmer and a PC, to program the board	first use only
SIM card	any SIM card, 1 piece, nano SIM format

The SIM card stays at the test station. It is used for every module.

2. Power on the board



1. Connect 12 V to the power connector: + and – as marked on the board.
2. All LEDs light up briefly, then **Busy** starts blinking.

If they do not, the board still has to be programmed once: see *Programming the test board*. After that the board works on its own, with no PC.

Leave the 12 V connected all day. The **Reset** button starts each new test.

3. Run the test

1. Wait until **Busy** stops blinking.
2. Put the SIM card in the modem, and the modem in the slot.
3. Press **Reset**. **Busy** starts blinking.
4. Wait until **Busy** stops blinking, which takes about 20 seconds.

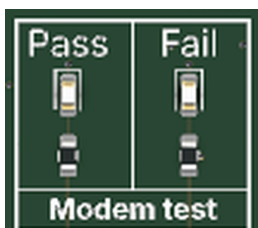


Busy blinking: the test is running. Do not read the result, and do not remove the modem.

Error blinking: the test board itself is broken. Its results cannot be trusted. Set the board aside and tell the engineer.

4. Read the result

Look at the two LEDs marked **Modem test**, next to the modem slot.



Pass on, does not blink: the modem is good.

Fail blinking: the modem is rejected. Count the flashes and write the number down.

The **Fail** LED gives a group of flashes, then a pause, then the same group again. Count the flashes in one group, and count twice to be sure. The smallest number is 2: there is never a single flash.

Flashes	Meaning
2	Modem does not answer
3	SIM card not detected
4	Modem uses too much power
5	No modem detected (no power consumption)
6	Wrong modem type

To test the next modem: wait until **Busy** has stopped blinking, swap the modem, and press **Reset**.